



Expression profiling of luminal B breast tumor in Indian women

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Abstract

Purpose In this study, we aimed at profiling of luminal B breast cancer specific gene expression pattern in Indian women using mRNA-seq and validation based on TCGA expression data.

Methods RNA isolated from luminal B tumor and adjacent normal tissues was used for library construction and sequencing. Reference-based assemblies of these reads were used for differential gene expression analysis using DeSeq2. The DEGs were evaluated using TCGA expression data. Kaplan–Meier survival method was used to evaluate association between genes showing luminal B specific differential expression pattern and breast cancer prognosis and statistical significance was assessed using log-rank test. Alternate splicing analysis was done using rmat.

Results Differential expression analysis identified 2371 differentially expressed genes (DEGs) in luminal B breast tumors in comparison with adjacent normal tissues of Indian Women. Of them, 1692 DEGs were validated using TCGA luminal B paired samples. Integration of this data with the DEGs obtained by comparative analysis of unpaired luminal B with luminal A unpaired samples from TCGA resulted in 291 DEGs showing luminal B specific expression pattern. Further, 26 genes of prognostic value were identified. Differential splicing analysis between luminal B tumors and adjacent normal tissues in our cohort led to the identification of 687 genes showing significant differential alternate splicing events.

Conclusion This study profiled gene expression pattern of luminal B tumors of Indian women and identified 26 key genes of prognostic value for luminal B breast cancer. This study also profiled differential alternate splicing and identified important alternate splicing events in luminal B breast cancer.

Keywords Transcriptome · Indian women · Breast cancer · Expression profiling · Luminal B · Prognostic genes · Differential alternate splicing

Abbreviations

DEGs	Differentially expressed genes	RI	Retention intron
PCA	Principal component analysis	SD	Standard deviation
TCGA	The Cancer Genome Atlas	PPAR	Peroxisome proliferator-activated receptor
OU	Osmania University	AMPK	AMP-activated protein kinase
RIN	RNA integrity number	PC1	Principal component1
FDR	False discovery rate	PC2	Principal component2
LFC	Log fold change	A3SS	Alternative 3' splice site
KEGG	Kyoto Encyclopaedia of Genes and Genomes	A3SS	Alternative 5' splice site
GO	Gene ontology	MXE	Mutually exclusive exons
TNBC	Triple negative breast cancer	IDC	Invasive Ductal Carcinoma
SE	Skipped exon	CAST	Calpastatin
		C-to-U	Cytidine to uridine
		ER	Estrogen receptor
		SAM	Sequence Alignment Map
		BAM	Binary Alignment Map
		PAM50	Prediction Analysis of Microarray 50
		PR	Progesterone receptor
		TNM	Tumor-node-metastasis
		ERBB2	Erb-b2 receptor tyrosine kinase 2

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Introduction

Breast Cancer, the disease of mammary epithelium, is one of the most commonly diagnosed cancers among Indian women, and a majority of patients present with locally advanced disease (Mathur et al. 2020; Kulothungan et al. 2022; Srivastava and Kumar 2022). Most widely accepted classification of breast cancer includes Her2-enriched, basal-like, luminal A and B breast cancers (Goldhirsch et al. 2011).

Luminal B tumors show higher expression of proliferation-related genes, and ER or PR receptors are expressed weakly (Ades et al. 2014). Luminal B tumors were correlated with higher grade and positive lymph node involvement. Luminal B breast cancers have a low survival rate in comparison with luminal A subtype (Cheang et al. 2009). In postmenopausal breast cancer patients with luminal B subtype, tumor is highly proliferative and the risk for distant metastatic disease is higher in first 5 years and is reduced later (Yu et al. 2019).

The high heterogeneity of luminal breast tumors influences diagnosis and treatment (Puppe et al. 2020). Extensive transcriptomic analysis is required to decipher this heterogeneity. RNA-seq (High throughput mRNA sequencing) is the standard method for analysing gene expression levels related to traits in higher eukaryotes. In order to understand phenotypic variations, it is essential to accurately identify DEGs between specific conditions (Costa-Silva et al. 2017). Different subtypes and stages are accompanied by a corresponding diversity in gene expression patterns that could be captured by RNA-seq (Chen et al. 2020; Yuan et al. 2019; Deva Magendhra Rao et al. 2019). Compared to other parts of the world only few papers have been published on the expression profiling of Indian breast tumors and no specific study is available on luminal B tumors of Indian women. In this study, we report differential gene expression pattern between luminal B tumor tissue and surrounding normal tissue in infiltrating ductal breast carcinoma of Indian women using mRNA-seq. By integrating expression data from our cohort and TCGA we detected genes showing luminal B specific differential expression pattern when compared to luminal-A breast cancer for identification of genes of prognostic value for luminal B breast cancer.

Materials and methods

Sample collection

Breast cancer tissue and adjacent normal tissue were obtained from four breast cancer patients at Mehdi Nawaj Jung (MNJ) Institute of Oncology & Regional Cancer Centre, Hyderabad, Telangana State, INDIA. The breast tumor samples were of luminal B subtype. Tissue samples collected and procedures followed in the current study were approved by the Ethical standing committee. Written informed consent was obtained to collect and use the tissue, from each patient, before collecting the samples. Tissue samples were snap frozen in liquid nitrogen, immediately after collection. The tissue samples were transferred to RNAlater solution and stored frozen at -80°C until further use.

RNA isolation, library preparation, and sequencing

RNA was isolated from these tissue samples by using RNeasy mini kit (Qiagen). RNA concentration and purity of samples were estimated using Nanodrop Spectrophotometer and Qubit Fluorometer. RNA quality was assessed by 2100 Bioanalyzer (Agilent Technology). Only samples with high quality ($\text{RIN} \geq 7.0$) were used for library construction. RNA sequencing libraries were prepared using poly A method with Illumina-compatible NEBNext[®] UltraTM Directional RNA Library Prep Kit (New England BioLabs, MA, USA). Illumina-compatible sequencing libraries were quantified by Qubit fluorometer (Thermo Fisher Scientific, MA, USA) and the fragment size distribution were analysed on Agilent 2200 TapeStation. Technical replicates of libraries were included for both normal and tumor samples. Sequencing was done using HiSeq X Ten platform with Paired end (150 bp) mode at Genotypic technologies, Bangalore. Apart from this, expression data and clinical data for 479 infiltrating ductal breast carcinomas were retrieved from TCGA (Genomic Data Commons Data Portal. <https://portal.gdc.cancer.gov/>). The PAM50 subtype classification data was obtained from cBioPortal (cBioportal. <https://www.cbioportal.org/>). These samples are here after referred to as TCGA cohort and samples from our study are referred to as OU cohort.

Table 1 OU cohort patient characteristics

S. no	Age	TYPE	TNM staging	ER	PR	HER-2
1	35	IDC	T4a/4b N1 M0	+	+	+
2	50	IDC	T3 N1 M0	+	+	+
3	49	IDC	T3 N1 M0	+	+	+
4	56	IDC	T3 N1 M0	+	–	+

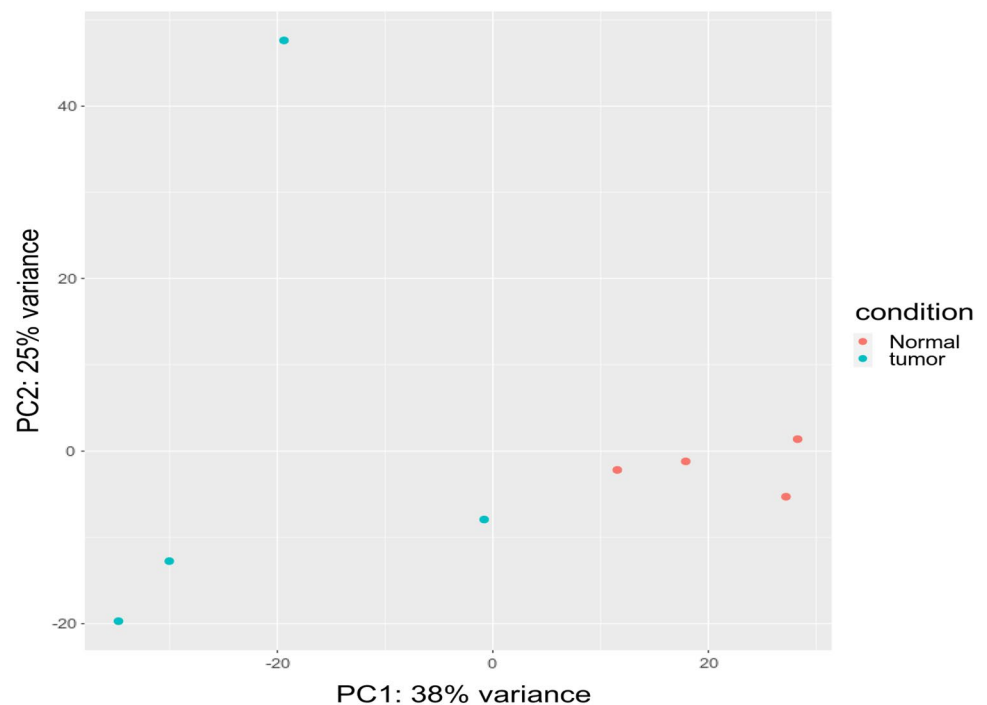
PR Progesterone receptor, TNM Tumor–node–metastasis, IDC Invasive Ductal Carcinoma, ER Estrogen receptor

Table 2 Details of the number of technical replicates used for tumor and adjacent normal tissue for each patient and the reads data summary for each sample

Sample	Condition	Technical replicates	No. of paired-end reads
1	Tumor	1	24,228,738
		2	25,515,574
		3	25,154,950
	Normal	1	26,656,173
		2	23,211,892
2	Tumor	1	17,996,822
		2	21,884,609
	Normal	1	25,437,826
		2	28,136,382
3	Tumor	1	25,754,975
		2	22,392,867
		3	24,017,311
	Normal	1	27,721,516
		2	27,192,593
4	Tumor	1	24,334,205
	Normal	1	24,540,235
		2	22,047,625

Table 3 Clinical information of TCGA samples

	Luminal B (n = 168)	Luminal A (n = 311)
Age		
< 60	89	164
> 60	79	147
Range	27–90	26–89
Sex (n%)		
Female	100 (100%)	100 (100%)
Male	0 (0%)	0 (0%)
Stage (n%)		
1	18 (10.71%)	82 (26.37%)
2	91 (54.17%)	163 (52.41%)
3	53 (31.55%)	52 (16.72%)
4	4 (2.38%)	6 (1.93%)
N/A	2 (1.19%)	11 (3.54%)
T (n%)		
T1	30 (17.86%)	119 (38.26%)
T2	109 (64.88%)	163 (52.41%)
T3	20 (11.90%)	16 (5.14%)
T4	9 (5.36%)	12 (3.86%)
TX	0 (0%)	1 (0.32%)
N (n%)		
N1	62 (36.9%)	144 (46.3%)
N2	63 (37.5%)	116 (37.3%)
N3	32 (19.05%)	31 (9.97%)
N4	9 (5.36%)	14 (4.5%)
NX	2 (1.19%)	6 (1.93%)

Fig. 1 Principal component analysis (PCA) of breast tumor and surrounding normal samples used in this study. The PC1 captured 38% of the total variance and PC2 captured 25% of the total variance. Each dot represents one sample

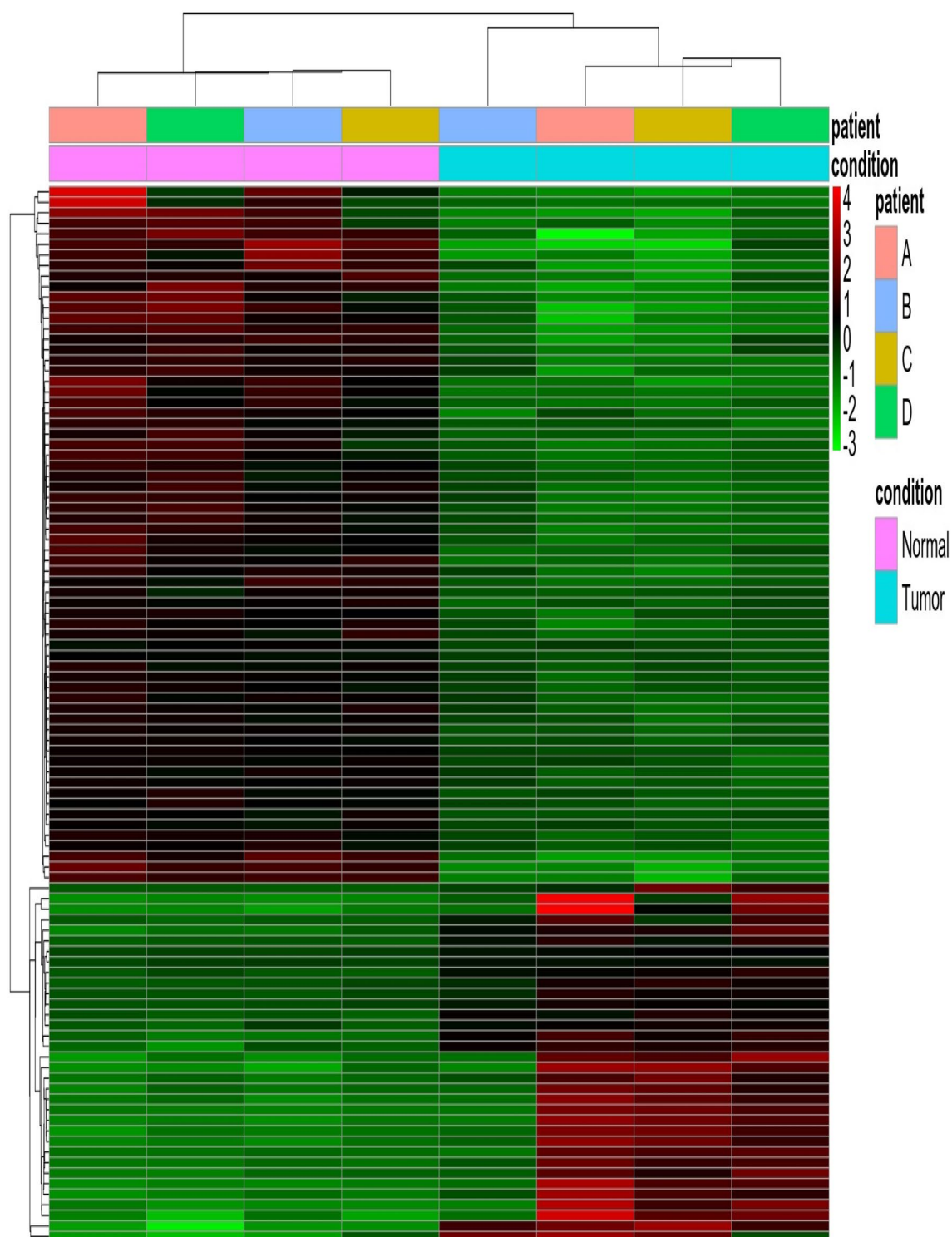


Fig. 2 Hierarchical clustering heatmap for top 100 differentially expressed genes (DEGs) in luminal B breast cancer. The hierarchical clustering of samples and genes was based on Euclidean distance.

The columns represent samples and the rows represent differentially expressed genes. The colour scale represents expression level

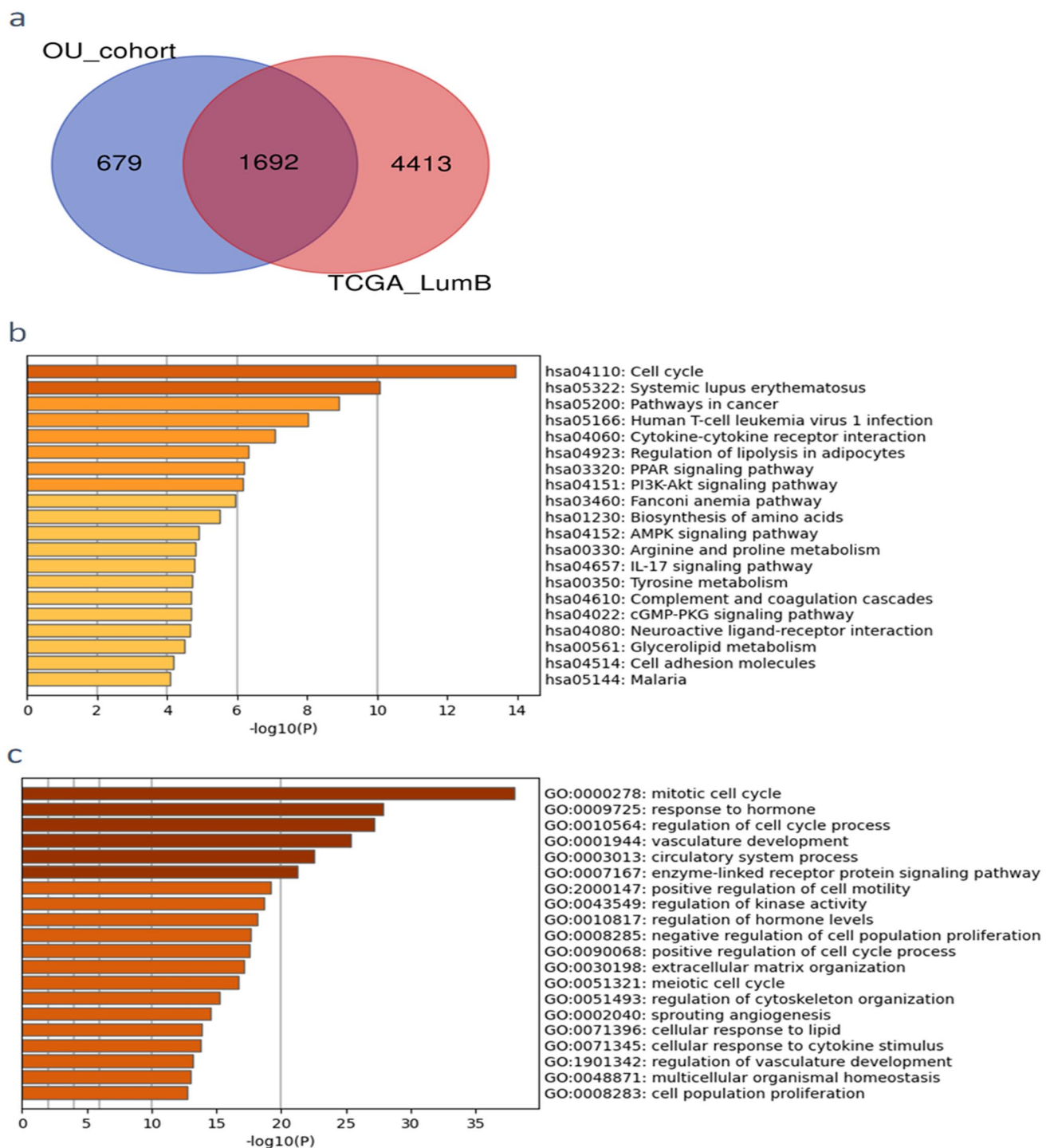


Fig. 3 Functional enrichment analysis of 1692 validated genes between OU cohort and paired Luminal B samples of TCGA using a FDR < 0.05 as cut off. **a** Venn diagram showing genes overlap-

ping between the OU cohort and paired luminal B samples of TCGA cohort, **b** significantly enriched KEGG pathways, **c** significantly enriched GO terms

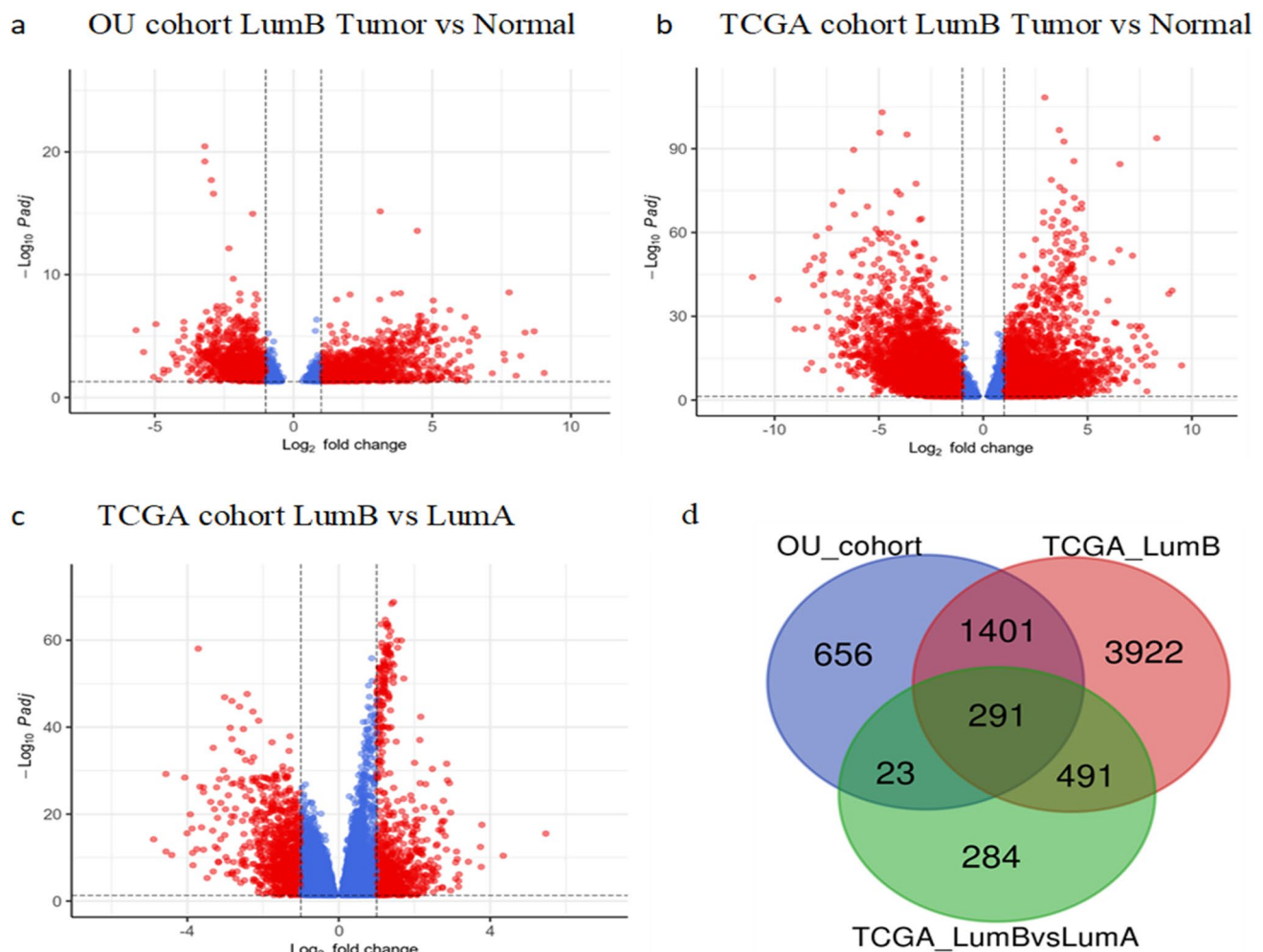


Fig. 4 Volcano plots representing differential expression pattern **a** OU cohort (Luminal-B tumor vs adjacent normal), **b** TCGA cohort (Luminal-B tumor vs adjacent normal), **c** LumB vs LumA. Red colour represents genes with log2 Fold Change ≥ 1 . Blue colour repre-

sents genes with log2 Fold Change ≤ 1 . **d** Venn plot showing 291 genes overlapping among luminal-B paired samples of OU cohort, luminal-B paired samples of TCGA cohort and TCGA LumB vs LumA

Pre-processing and reference-based assembly

The raw reads in fastq format were quality checked by using FastQC (Version 0.11.9) (Andrews 2010). Adapters were removed using Cutadapt version 4.2 (Martin 2011). Reference-based assembly was done using Hisat2, splice aware aligner (Kim et al. 2019). The human reference genome GRCh38.p13, downloaded from Ensembl (Ensembl. http://asia.ensembl.org/Homo_sapiens/Info/Index), was indexed and used in the assembly. The sequence alignment data (SAM) files obtained after the assembly were converted into binary alignment file (BAM) files using Samtools (version 1.12). These BAM files were indexed and sorted before use (Li et al. 2009). The BAM files of technical replicates were merged using Samtools-merge.

Differential expression analysis

The read mapping results were summarized for each gene, annotated using gene annotation file for the reference genome GRCh38.p13. Only coding genes were annotated. Feature Count of Rsubread package in R was used to generate read count summary (Liao et al. 2014). Read pairs with both the ends mapped properly were only counted. DESeq2 was used for differential expression analysis (Love et al. 2014). Differential expression analysis between tumor and corresponding surrounding tissues was estimated as log fold change (logFC). The statistical significance of differential expression estimations was assessed by wald test and corrected for by Benjamini & Hochberg method. Genes that were differentially expressed with p -value < 0.05 were considered as significant. PCA analysis was done to see if

samples clustered together. Hierarchically clustered heatmap was generated with p -value < 0.05 and $|\log\text{FC}| \geq 1$ using pheatmap in R.

Identification of luminal B specific expression pattern

The read count data of TCGA cohort (starcounts) were loaded into DESeq2 for analysis. DEGs were obtained by comparing expression profiles of paired luminal B tumor and adjacent normal samples in TCGA cohort. This was used for validation of DEGs from OU cohort data obtained in the previous step (Fig. 3). DEGs identified by integration DEGs from OU cohort (Tumor vs. Adjacent normal) and TCGA Paired luminal B samples (Tumor vs. Adjacent normal) with DEGs obtained by comparing expression profiles of TCGA unpaired luminal B and Luminal A samples was defined as genes showing luminal B specific expression pattern (Fig. 4). The genes with $|\log\text{FC}| \geq 1$ and p -value < 0.05 were considered significant.

Functional analysis

Enrichment analysis was done using Metascapev3.5.20230101 (Zhou et al. 2019a, b). The genes were enriched for GO terms and KEGG pathway ($p < 0.01$). The p -values were corrected for false discovery rate by Benjamini & Hochberg method.

Statistical analysis

The association between genes showing luminal B specific differential expression pattern and breast cancer prognosis was evaluated using Kaplan–Meier survival method and Univariate Cox proportional hazards model analysis using “survival” package and “survminer” package in R. Samples from TCGA with clinical data available were included in the survival analysis. Samples were categorized into two groups based on expression levels: (1) High expression group (with expression values above the median) and (2) Low expression group (with expression values below the median). The Kaplan–Meier survival method was used to estimate the survival probabilities of High expression and low expression groups and log-rank test was used to assess the statistical significance of difference in survival probabilities between the groups. All differences were considered statistically significant if the p -value was < 0.05 .

Alternate splicing analysis

Alternate splicing analysis was done using the event-based tool rmat (Shen et al. 2014). We used rmat to identify skipped exon (SE), alternative 5' splice site (A5SS),

alternative 3' splice site (A3SS), mutually exclusive exons (MXE), and retained intron (RI) events in OU cohort. Significant differential alternate splicing events between tumor and adjacent normal tissue were filtered using maser in R and were visualised using rmat2sashmplots.

Results

Data used in the analysis

Reads generated by sequencing four luminal B tumor samples of Indian women and corresponding adjacent normal tissues were used in this analysis (Table 1). Multiple technical replicates were sequenced and used for both tumor and adjacent normal tissues. The number of reads in tumor samples ranged from 17,996,822 to 25,754,975, while the number of reads in the adjacent normal samples ranged from 23,211,892 to 28,136,382 (Table 2). These paired-end reads were quality checked using FastQC and adapters were removed using Cutadapt before use in differential expression analysis. Towards identification of genes showing luminal B specific expression pattern TCGA expression data from 483 infiltrating ductal Breast carcinoma samples and 17 paired normal sample expression data were integrated with OU cohort data in the expression analysis. The TCGA cohort includes expression data from 315 luminal-A (LumA) samples and 168 luminal B (LumB) samples. Additionally, expression data for adjacent normal tissues of 17 out of 168 luminal B type were also included (Table 3).

Reference-based assembly

For expression analysis, reference-based assembly of reads from tumor and adjacent normal samples, including their replicates, were assembled on to human reference genome (GRCh38). The reference genome was indexed and the reads were assembled on to it using Hisat2. Uniquely aligned reads were only used in the analysis and the alignment rate ranged from 85 to 91%. The bam files of technical replicates of each sample were merged together using Samtools before read count.

Differential expression analysis

The merged bam files were used for read count using feature counts. Differential expression analysis of the read count data using DESeq2 identified a total of 2371 significant DEGs ($\text{padj} < 0.05$ and $\text{LFC} \geq 1$) between tumor and corresponding adjacent normal samples of OU cohort, of them 1196 genes were found to be downregulated, while

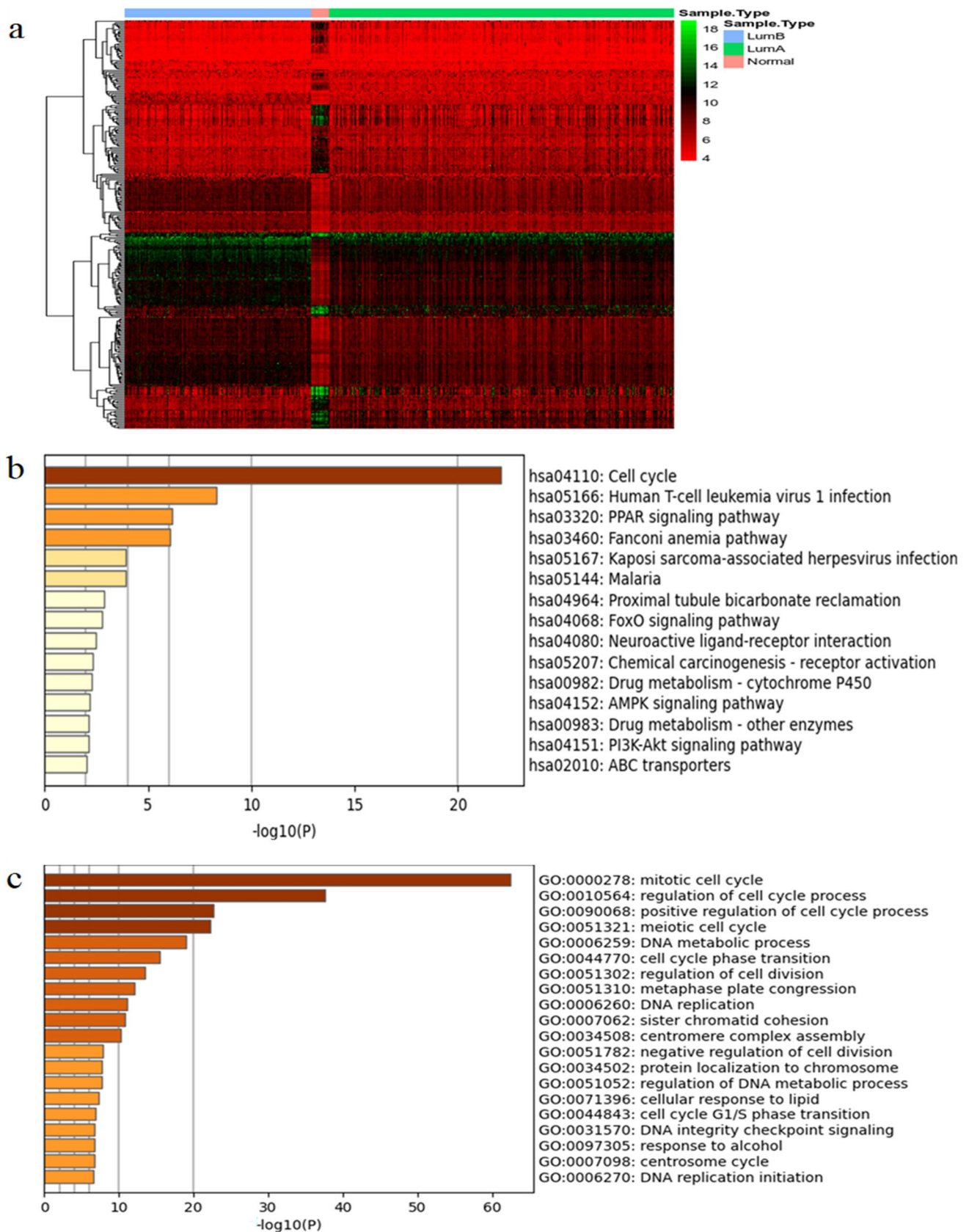


Fig. 5 Luminal B specific expression pattern **a** Heatmap showing log2 transformed read counts of TCGA LumB ($n=168$), adjacent normal samples ($n=17$) and LumA ($n=311$) of 291 genes showing luminal B specific expression pattern. Functional enrichment of 291 DEGs showing significantly enriched, **b** KEGG pathways and **c** GO terms ($p<0.01$)

1175 genes were found to be upregulated (Supplementary file 1). PCA analysis carried out on the log2 transformed gene expression data of OU cohort indicated 38% variance between tumor and normal samples indicating clear difference between the transcriptome profiles of tumor and normal tissue samples ($PC1=38\%$, Fig. 1). Hierarchical clustering heatmap of top 100 upregulated and downregulated genes, generated using pheatmap, is shown in Fig. 2. Differential expression analysis between 17 Luminal B tumor and paired normal samples from TCGA cohort identified 6105 DEGs (Supplementary file 2). Of them 1692 genes were common between OU cohort and TCGA luminal B paired samples (Fig. 3a, Supplementary file 3).

Functional analysis

Functional enrichment analysis for 1692 DEGs obtained in the above step was carried out using clusterProfiler. Using a padj cut off of 0.05, genes were enriched for KEGG pathways and GO terms (Fig. 3b, c). The genes were enriched for key cancer related KEGG pathways like Cell cycle, PPAR signalling pathway, regulation of lipolysis in adipocytes, Cytokine-cytokine receptor interaction, Fanconi anaemia pathway and AMPK signalling pathway.

Luminal B specific expression pattern

Comparing the expression data of 168 unpaired luminal B samples with 311 luminal A samples resulted in identification of 1089 DEGs between luminal B and luminal A subtypes. (Supplementary file 4). By integration of this data with differentially expressed data of OU cohort (Tumor vs. Adjacent normal) and TCGA luminal B paired samples (Tumor vs. Adjacent normal) we identified 291 genes showing luminal B specific differential expression pattern (Supplementary file 5). These genes showed relatively higher or lower mean expression values in luminal B samples in comparison with luminal A samples. Volcano plots for all the comparisons are shown in Fig. 4. Heatmap showing the log2 transformed read counts of genes showing luminal B specific differential expression pattern and functional enrichment analysis are shown in Fig. 5.

Table 4 Base-line and follow-up information of the 168 Luminal B patients (TCGA) in survival analysis

	Statistics	N
Age (years)		
Mean $\pm$ SD	58.84 $\pm$ 13.27	168
Range (min–max)	27–90	0
Sex (%)		
Female	100%	168
Vital status		
Alive	84.52%	142
Dead	15.48%	26
Median follow-up time (days)	640	
Ajcc stage (%)		
1	10.71%	18
2	54.17%	91
3	31.55%	53
4	2.38%	4
N/A	1.19%	2

Ajcc American Joint Committee on Cancer, N/A not available

Evaluation of prognostic value

To analyse prognostic potential of the DEGs showing luminal B specific differential expression pattern using survival package in R. The Base-line and follow-up information used for this analysis is shown in Table 4. We identified 26 genes that are significantly associated with overall survival of luminal B breast cancer patients. In our study, overexpression of EXO1, CENPI, ANLN, ERCC6L, DTL, TPX2, E2F7, and APOBEC3B was found to be associated with poor overall survival in luminal B breast cancer and down-regulation of genes TMEM132C, PLIN1, FABP4, KCNIP2, SCARA5, C7, C14orf180, S100B, TNXB, CTSG, ADIPOQ, CIDEA, TNMD, ABCA10, GDF10, HEPACAM, CHRDL1, and ABCA8 was found to be associated with adverse outcomes (Figs. 6, 7).

Differential splicing analysis in luminal B breast cancer subtype

Differential splicing analysis between tumor and adjacent normal tissues in OU cohort led to the identification of 1097 known and novel significant differential splicing events in 687 genes including 621 skipped exon (SE), 189 intron retention (RI), 114 alternative 3' splice site (A3SS), 82 mutually exclusive exon (MXE) and 91 alternative 5' splice site (A5SS) events using an average read cut off of 25, p-value cut off of 0.05 and Inclusion Level difference cut

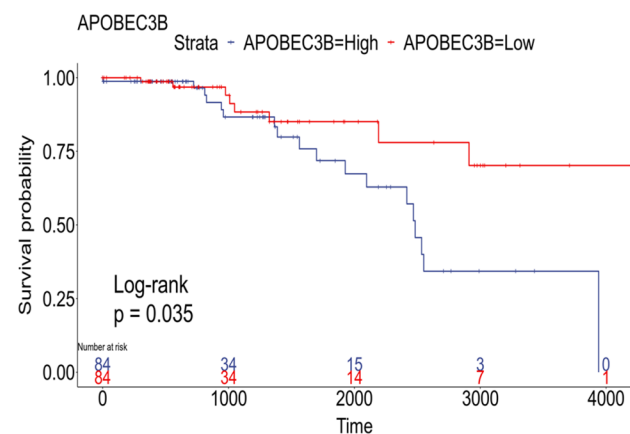
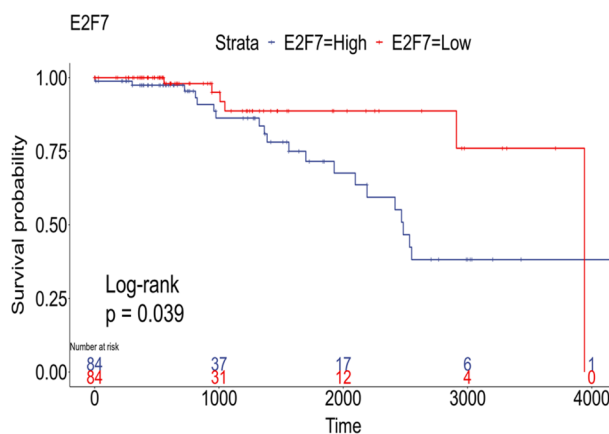
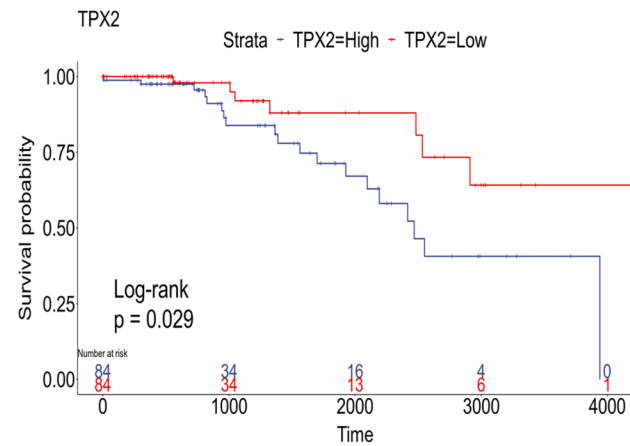
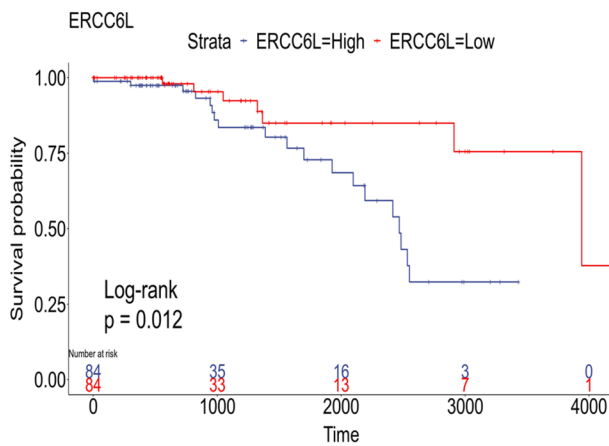
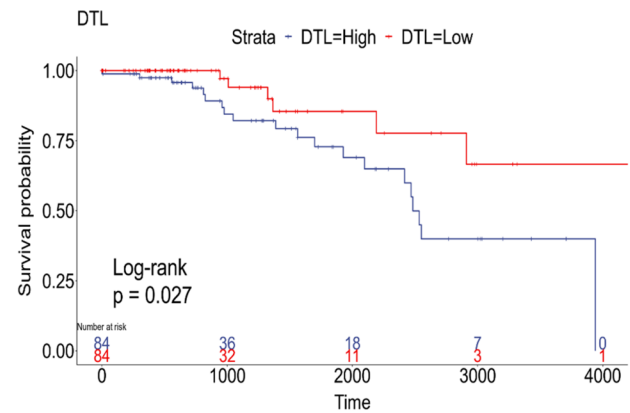
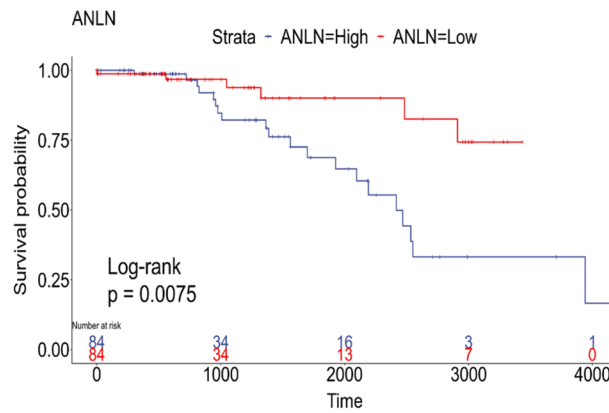
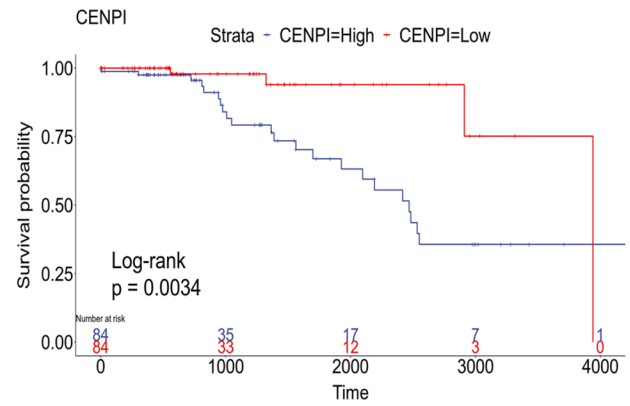
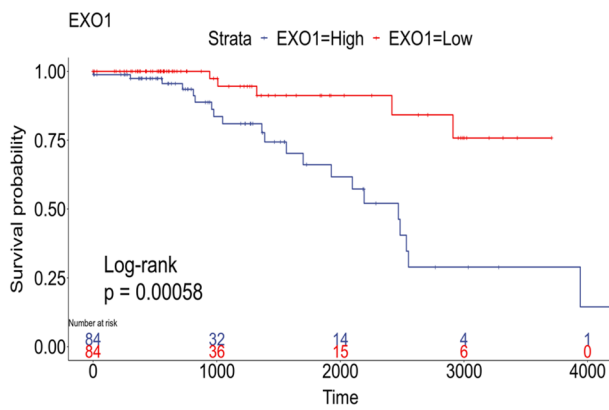


Fig. 6 Association between overexpressed genes in luminal B breast cancer and overall survival. Survival curves of 8 significantly associated survival genes among 291 genes showing luminal B specific expression pattern. Statistical significance was assessed by log-rank test

off of 0.2 (Supplementary file 6). The global splicing levels of different splicing events are shown in Fig. 8. Differentially expressed genes that were also differentially spliced were obtained by integrating the significant DEG data with that of significant splicing events obtained. Using this method, we identified 61 differentially spliced genes that were also differentially expressed. We found 80SE, 8 A5SS, 8 A3SS, 9MXE, and 17 RI events in 61 genes. The top differential splicing events identified based on p-value, inclusion levels and read coverage was shown by gene Calpastatin (CAST). In CAST we observed increased skipping of exon 3 in tumor samples (chr5: 96,726,794–96,726,859, Fig. 9a). Among the 26 prognostic genes we have identified ABCA10 as differentially spliced. In ABCA10 we observed decreased skipping of exon in tumor samples (chr17: 69,178,920–69,179,008, Fig. 9b). Among the 687 genes showing differential splicing events, 80 genes were found to be differentially spliced in breast cancer samples in a study reporting cancer specific alternate splicing (Danan-Gotthold et al. 2015). The difference in percent spliced in was in the same direction for all the identified genes. These events are considered to be highly significant. GOLGA4, PRMT2 and KIF13A were three genes differentially spliced in our data that were reported to be differentially spliced also in non-small cell lung cancer, in a separate study (de Miguel et al. 2016). This shows that 83 genes accurately reflect the alternate splicing changes in breast cancer.

Discussion

Early gene expression studies have shown that ER+ breast cancer is made of two distinct diseases, luminal A and luminal B, with different biology. Among these two, luminal B cancers showed worse outcomes when compared to luminal A cancers and they have a distinctive molecular phenotype from that of luminal A breast cancers (Creighton et al. 2012). In this work we made expression profiling of Luminal B breast tumors of Indian women in comparison with corresponding normal tissues. We detected genes showing luminal B specific differential expression pattern when compared to luminal-A breast cancer for identification of genes of prognostic value for luminal B breast cancer by integrating expression data from our cohort and TCGA.

About 2371 significant DEGs were identified by comparative analysis of transcriptomes of luminal B breast tumors

of Indian women with their corresponding adjacent normal tissues ($p < 0.05$; LFC > 1;). Of them, 1692 DEGs were validated using TCGA luminal B paired samples. Integration of this data with the DEGs obtained by comparative analysis of unpaired luminal B with luminal A unpaired samples from TCGA resulted in 291 DEGs showing luminal B specific expression pattern.

This study identified 26 genes of prognostic value i.e., that are significantly associated with overall survival of luminal B breast cancer patients. Out of these, 8 genes were found to be upregulated in luminal B tumor tissue when compared to the corresponding normal tissue. These genes include EXO1, CENPI, ANLN, ERCC6L, DTL, TPX2, E2F7, and APOBEC3B. Higher expression of these genes predicted poor survival. Seven out of these eight genes are associated with DNA repair and cell cycle (Table 5).

Exonuclease 1 (EXO1) is a mismatch repair protein with 5' to 3' exonuclease and RNase H activity. EXO1 is strongly expressed in breast cancer tissue in comparison with the normal tissue. This over expression showed strong association with carcinogenesis and poor survival outcomes in breast cancer (Mimitou et al. 2008; Qi et al. 2019; Liu et al. 2021). Centromere proteins (CENPs) participate in the kinetochore assembly during mitosis (Fritzler et al. 2011). CENPI is required for timely progression through G₂ phase and mitosis. In our study we observed that luminal B breast tumors show relatively higher expression when compared to luminal A breast cancer and higher expression is associated with unfavourable prognosis in luminal B tumors. CENP-I has been reported to be present in ER+ but not in estrogen receptor-negative (ER-) breast carcinomas, high expression of CENPI in ER-positive breast cancer predicted relapse and poor survival. Consistent with our findings, Luminal A tumor was reported to show low expression of CENPI when compared to luminal B. Also, lower expression was found to be associated with favourable prognosis in luminal A tumors (Thangavelu et al. 2017; Liu and Liu 2022).

Anillin (ANLN) is an actin binding protein. It is essential for cytokinesis. Elevated expression of ANLN expression in breast cancer tissue is correlated to poor prognosis and was considered to be an independent prognostic marker in breast cancer (Xiao et al. 2022; Magnusson et al. 2016). Excision Repair Cross-Complementation group 6 Like (ERCC6L) is a DNA helicase. ERCC6L is found to be upregulated in 12 solid cancers including breast cancer. Higher ERCC6L expression was found to promote cancer cell proliferation (Xie et al. 2019; Pu et al. 2017).

Denticleless E3 ubiquitin protein ligase homolog (DTL) is involved in promoting breast cancer cell growth (Ueki et al. 2008). In breast cancer patients, upregulation of this gene is reported to be significantly associated with worse overall survival (Zeng et al. 2021). The microtubule associated

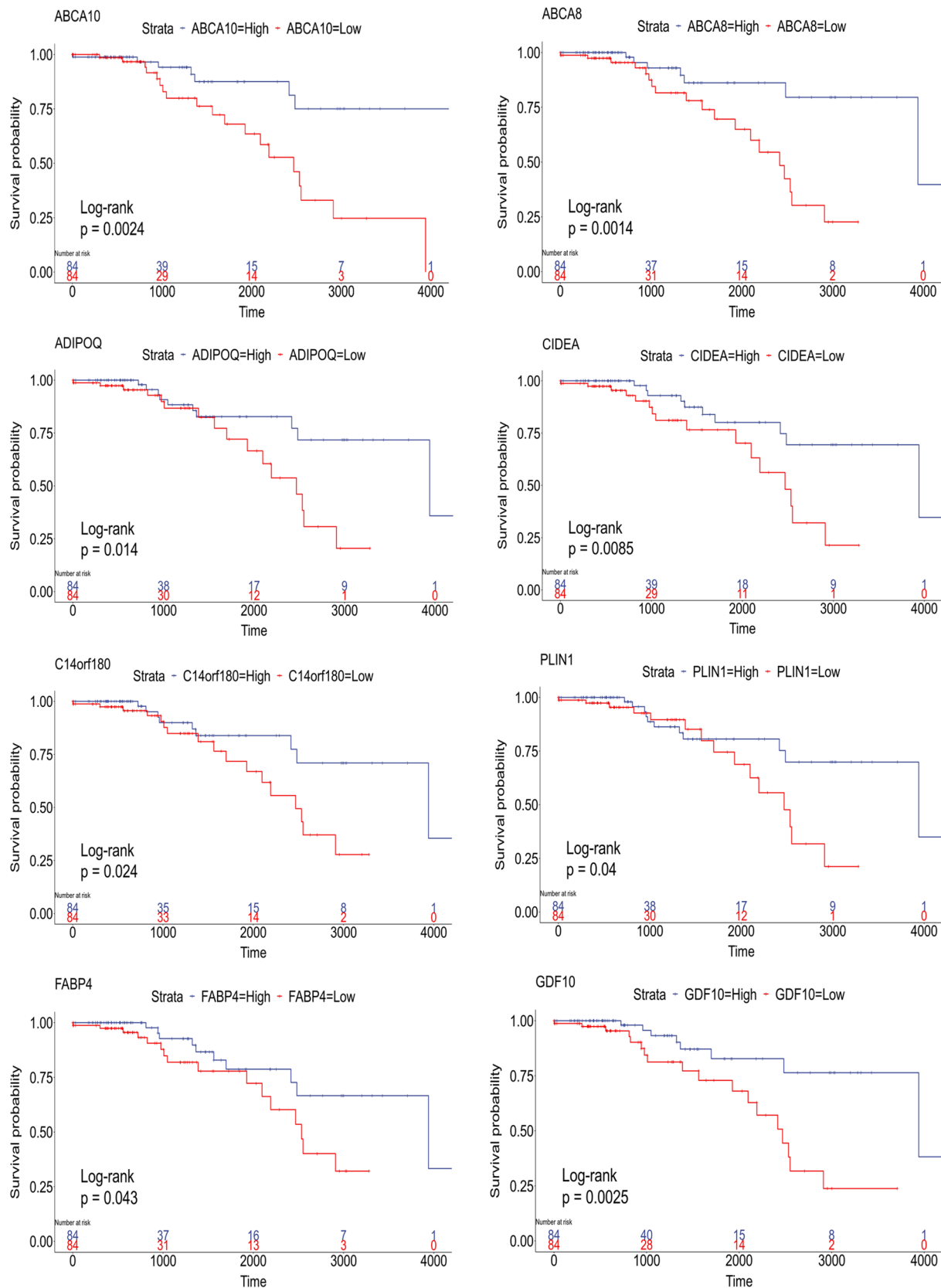


Fig. 7 Association between under-expressed genes in luminal B breast cancer and overall survival. Survival curves of 18 significantly associated survival genes among 291 genes showing luminal B specific expression pattern. Statistical significance was assessed by log-rank test

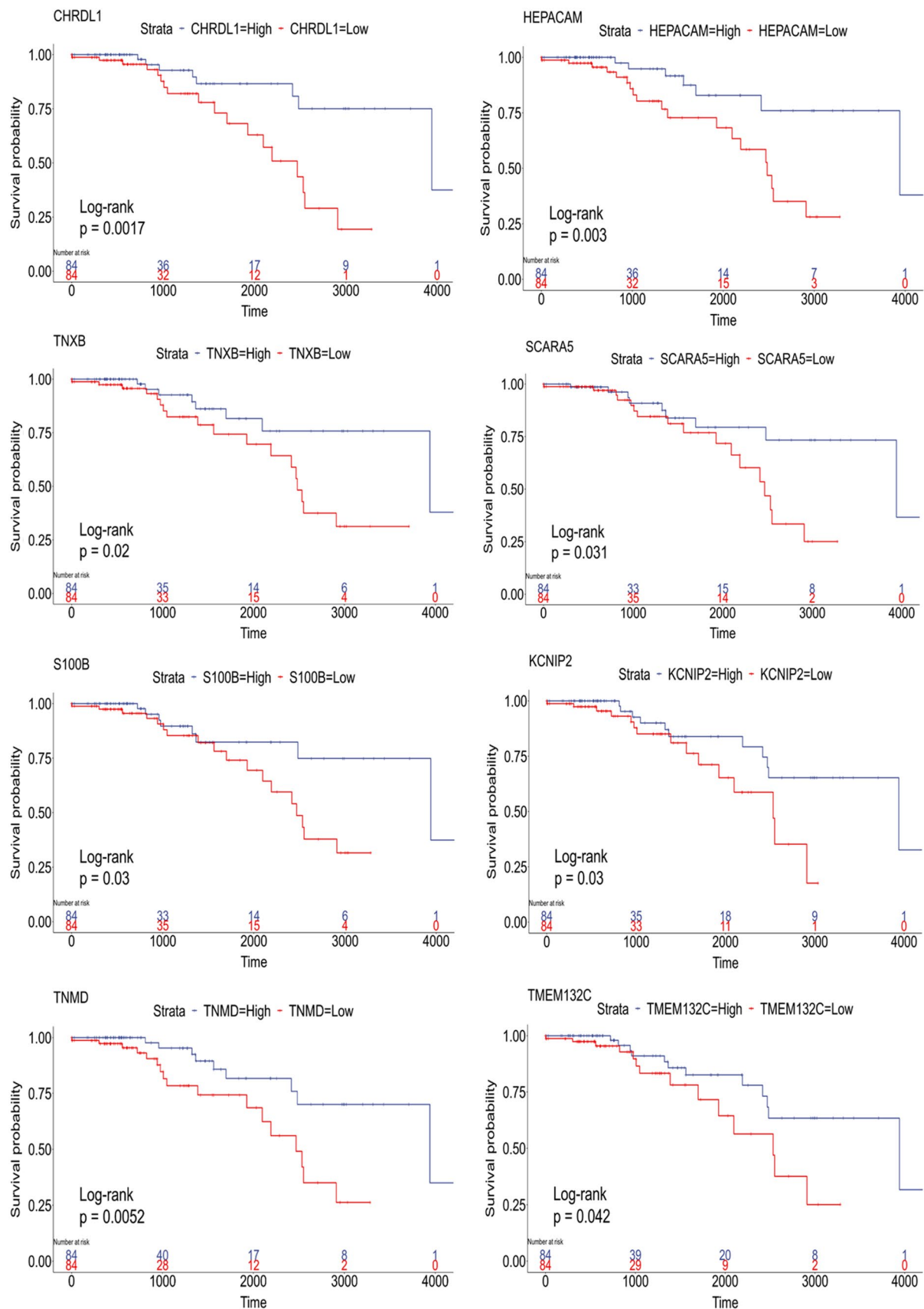


Fig. 7 (continued)

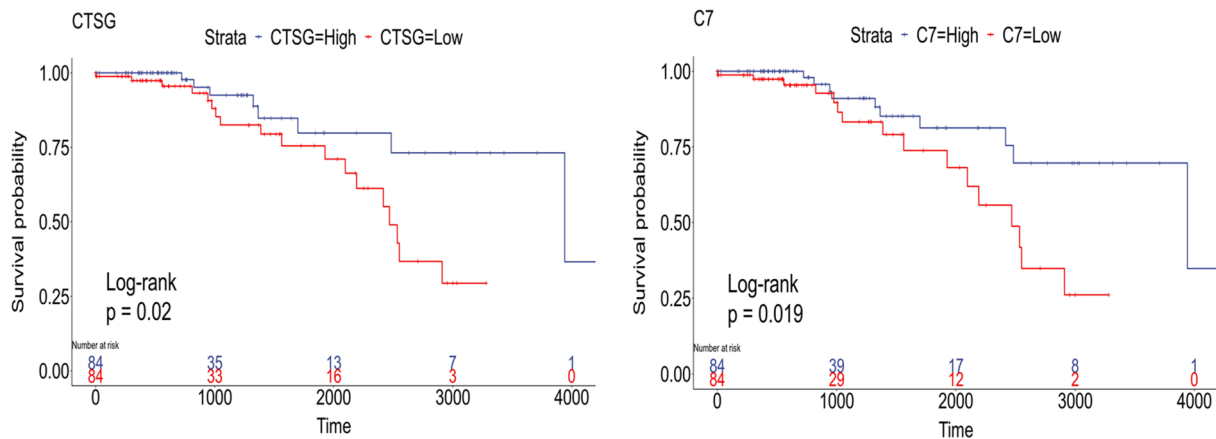


Fig. 7 (continued)

TPX2 protein has been found to be involved in DNA damage response and mitotic spindle assembly. Depletion of TPX2 in breast cancer cells was found to promote apoptosis and repress cell proliferation and migration. Higher expression of TPX2 predicts worse overall and disease specific survival in TNBC (Chen et al. 2018; Matson et al. 2021). Also, our study predicts a strong association between elevated expressions of TPX2 with poorer overall survival in luminal B tumors.

E2F Transcription Factor 7 (E2F7) regulates DNA repair and genomic integrity (Mitxelena et al. 2018). In breast cancer patients receiving tamoxifen treatment overexpression of E2F7 is linked with higher relapse rate and poor prognosis (Chu et al. 2015). Apolipoprotein B mRNA editing enzyme catalytic subunit 3B (APOBEC3B) is a cytosine deaminase that causes C-to-U transitions. It was found to be upregulated in many primary breast tumors and is an important mutational source in breast cancer (Burns et al. 2013). APOBEC3B mRNA expression was found to be associated with poor prognosis in Luminal A and Luminal B subtype (Mao et al. 2020). In this study we report mRNA expression of APOBEC3B is significantly correlated with worse overall survival of luminal B breast cancer patients.

Eighteen genes found to be downregulated in breast cancer tissue of our samples when compared to the corresponding normal tissue are ABCA10, ABCA8, ADIPOQ, C14orf180, C7, CHRDL1, CIDEA, CTSG, FABP4, GDF10, HEPACAM, KCNIP2, PLIN1, S100B, SCARA5, TMEM132C, TNMD, and TNXB. Lower expression of these genes is found to be associated with poor prognosis. Seven out of these eighteen genes are associated with lipid metabolism (Table 6).

ABCA10 and ABCA8 are members of the ABC transporter super family, they have been found to regulate lipid transport and homeostasis (Dvorak et al. 2017; Wenzel et al. 2003). Downregulation of ABCA10 in different

subgroups of breast cancer was found to be correlated with shorter overall survival (Chu et al. 2022). Overexpression of ABCA8 was found to inhibit breast cancer cell proliferation through the AMPK/mTOR signalling pathway (Ly et al. 2022). Adiponectin (ADIPOQ) is an adipocytokine secreted by adipocytes in the breast tumor microenvironment. It inhibits breast cancer cell growth by inducing of cytotoxic autophagy (Chung et al. 2017). In breast cancer, ADIPOQ downregulation has been predicted to show strong association with aggressive phenotypes (Llanos et al. 2021).

The gene, chromosome 14 open reading frame 180 (C14orf180), plays a role in differentiation of adipocytes and regulation of lipid storage (Kerr et al. 2019). It was found to be expressed at low levels in breast tumor than the adjacent normal tissue (Mamoor et al. 2021). Cell Death Inducing DFFA Like Effector A (CIDEA) promotes apoptosis and is involved in lipid metabolism. Expression of CIDEA was found to be downregulated in breast invasive carcinoma (Bortolotto et al. 2017; Gao et al. 2021). The fatty acid-binding protein 4 (FABP4) provides lipids from surrounding micro environments of cancer cells which are deprived of nutrients. This gene is downregulated in our samples.

Higher expression of Chordin-Like 1 (CHRDL1) inhibits breast cancer cell migration and invasion through inhibition of Bone Morphogenetic Protein (BMP) signalling. CHRDL1 was found to be overexpressed in luminal A and basal-like breast cancers when compared to luminal B or Her2 Positive breast cancer and over expression of CHRDL1 in breast tumor is found to show significant association with better clinical outcomes (Cyr-Depauw et al. 2016). Interestingly, in our study also we found expression of CHRDL1 to be higher in luminal A when compared to luminal B. Lower expression of CHRDL1 is found to be significantly correlated with poor prognosis. Growth differentiation factor-10 (GDF10) acts as a tumor suppressor by inhibiting proliferation and epithelial-mesenchymal transition in mammary epithelial

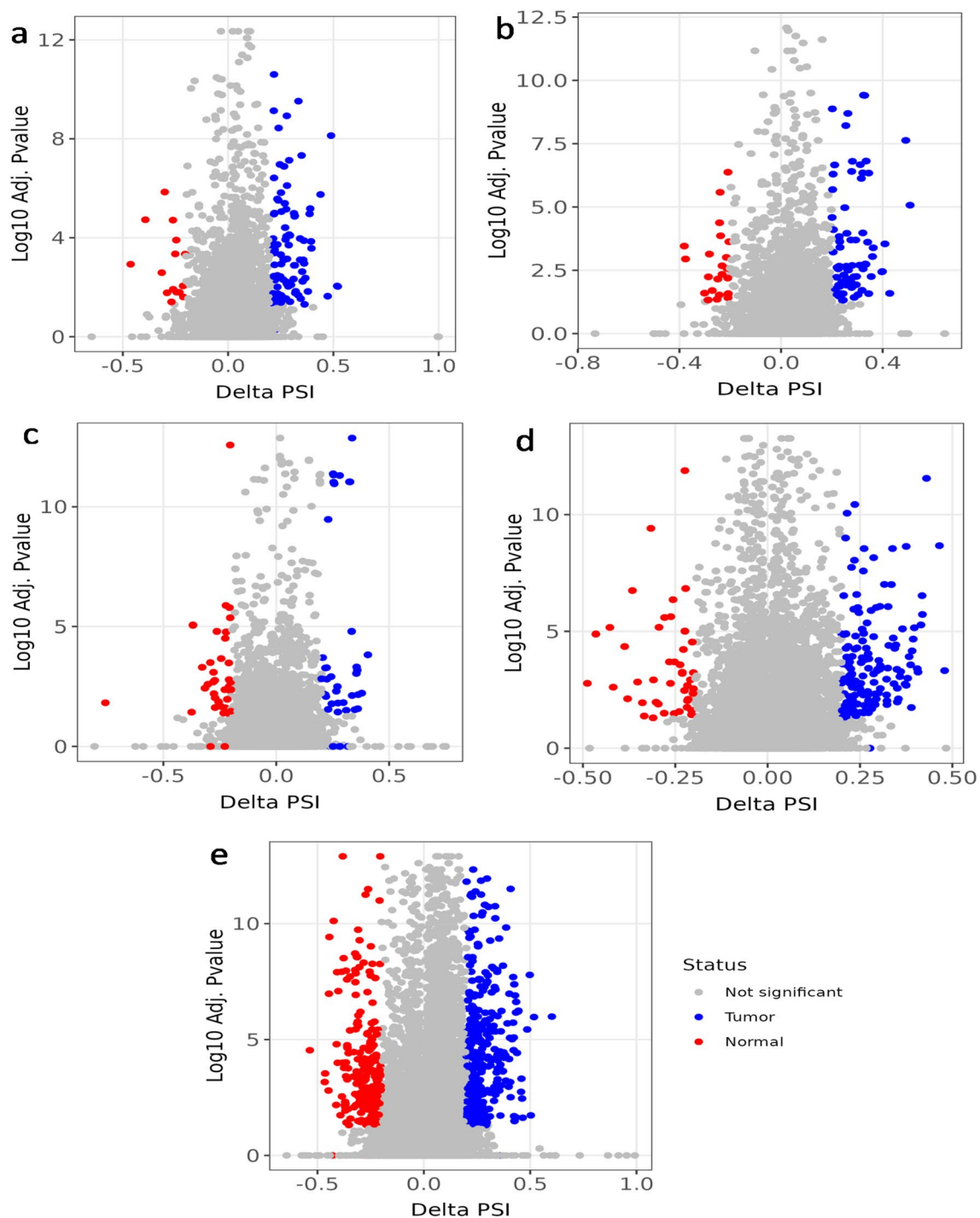


Fig. 8 Volcano plots of significant splicing events showing FDR levels and minimum change in PSI between luminal B tumor and adjacent normal tissue for each type of splicing event **a** A3SS event, **b**

A5SS event, **c** MXE event, **d** RI event and **e** SE event (FDR < 0.05 and Inclusion level difference > 0.2)

cells (Zhou et al. 2019a, b). Downregulation of GDF10 is positively correlated with lower survival rate in BC patients (Rahman et al. 2020). In breast cancer, expression of S100 Calcium Binding Protein B (S100B) was observed to be low. Higher expression of S100B predicted a better distant

metastases-free survival in all breast cancer patients and good overall survival in ER-negative breast cancer patients (Yen et al. 2018).

Complement Component 7 (C7) participates in the coordination of innate and adaptive immune reactions. Unlike

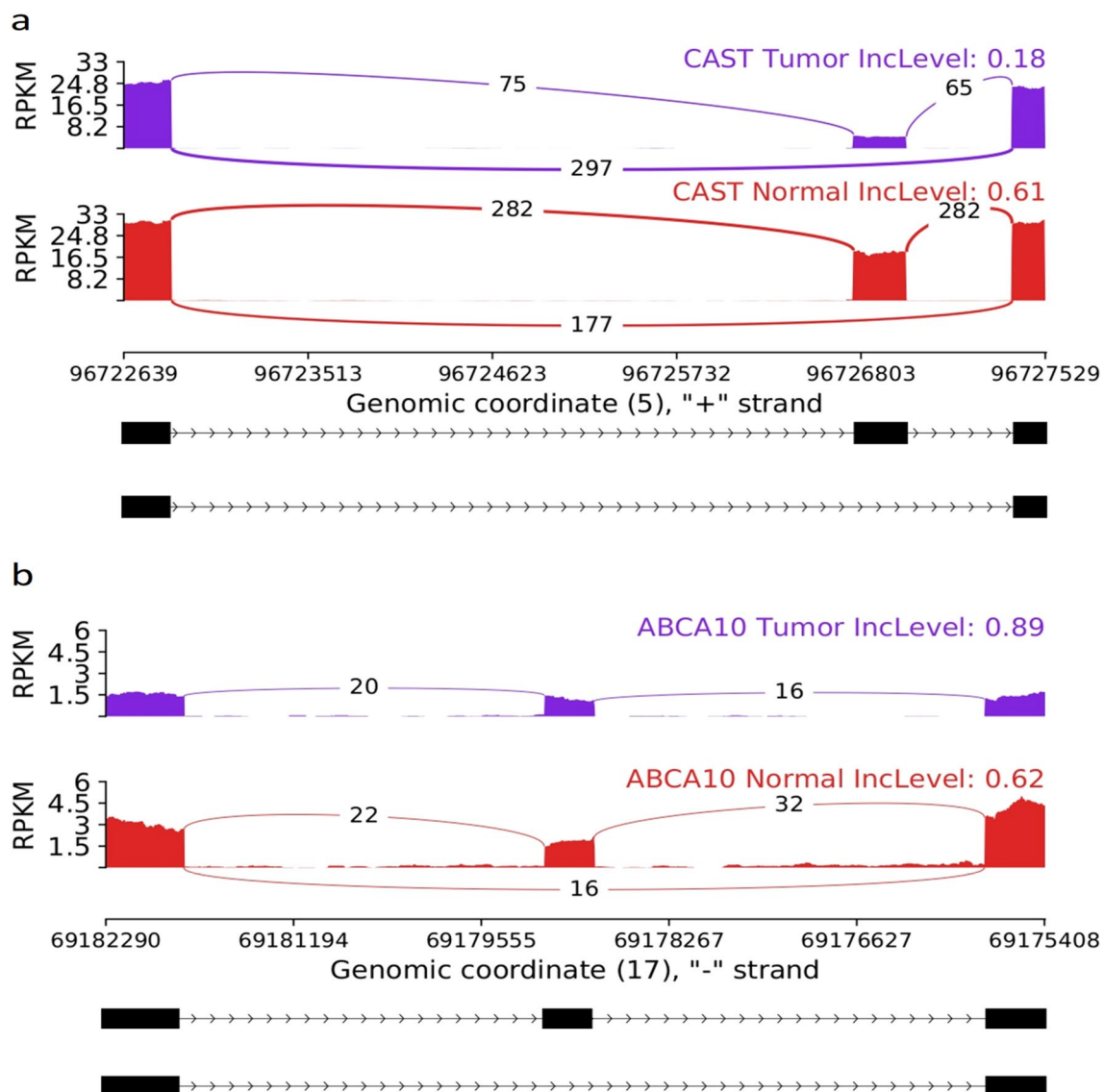


Fig. 9 Sashimi plots of selected differentially spliced genes. Gene ABCA10 shows an exon skipping event (chr17: 69,178,920–69,179,008) involving exon 23 and CAST shows exon skipping event involving exon 3 (chr5: 96,726,794–96,726,859). In each plot, the

numbers on the curves are junction read counts. The tumor samples are shown in purple while the normal samples are shown in red. The black bars represent the exons, while the lines represent the introns of the matching gene

Table 5 Upregulated genes of prognostic value associated with Cell cycle and DNA repair

S. no	Name of the gene	Expression in OU cohort and TCGA	Function
1	EXO1	Upregulated	DNA repair and recombination
2	CENPI	Upregulated	Regulation of chromosomal alignment and segregation
3	ANLN	Upregulated	Cytokinesis
4	DTL	Upregulated	DNA replication, DNA damage repair and cell cycle regulation
5	ERCC6L	Upregulated	Mitosis and chromatin remodelling
6	TPX2	Upregulated	Mitotic spindle formation
7	E2F7	Upregulated	Regulation of G1/S transition

Table 6 Downregulated genes of prognostic value associated with lipid metabolism

S. no	Name of the gene	Expression in OU cohort and TCGA	Function
1	ABCA10	Downregulated	Lipid homeostasis
2	ABCA8	Downregulated	Lipid transport
3	ADIPOQ	Downregulated	Lipid metabolism
4	PLIN1	Downregulated	Lipolysis and lipid storage
5	FABP4	Downregulated	Lipid metabolism
6	CIDEA	Downregulated	Lipid metabolism
7	C14orf180	Downregulated	Regulation of lipid storage

our samples upregulation of C7 in ductal carcinoma tissues predicted worse prognosis particularly in the luminal B and triple negative subtypes (Zhang et al. 2021). Cathepsin G (CTSG) has been found to be highly expressed in metastatic breast tumors (Tan et al. 2013). In our study the expression of Cathepsin G was observed to be down-regulated and predicted poor overall survival. Low expressing genes Hepatic And Glial Cell Adhesion Molecule (HEPACAM), Scavenger Receptor Class A Member 5 (SCARA5), Transmembrane Protein 132C (TMEM132C), Tenomodulin (TNMD), Tenascin (TNXB) and Kv channel-interacting protein 2 (KCINP2) in our study, were also reported to be downregulated in breast cancer (Moh et al. 2008; Ulker et al. 2018; Zhang et al. 2020; Deng et al. 2019; Liot et al. 2020; Dong et al. 2018).

Differential splicing analysis between tumor and surrounding normal tissues from OU cohort lead to the identification of differential splicing events in 687 genes. Approximately 9% of these 687 genes are differentially expressed between tumor and surrounding normal tissue. In the current study, increased skipping of Exon 3 of Calpastatin (CAST) has been observed in luminal B breast cancer. CAST is a protease inhibitor acting on calpain family. This protein has 4 conserved repetitive inhibitory domains and a regulatory N terminal domain. Exon 3 is present in the L domain which is a part of N terminal regulatory region of the gene. This exon encodes a lysine rich peptide sequence and it has been postulated that this sequence may act as a nuclear targeting signal (Lee et al. 1992; Sparatore et al. 2019). Downregulation of mRNA containing Exon 3 has been observed to be significantly correlated with lymphovascular invasion in breast cancer (Storr et al. 2011). ABCA10 is a cholesterol-responsive gene and plays role in lipid homeostasis. We found a decreased skipping of exon spanning region (chr17: 69,178,920–69,179,008) in tumor tissue in comparison with surrounding normal tissue, with 11% skipping in

tumor and 38% skipping in normal samples. The functional consequence of decreased skipping is unclear and needs further investigation.

Conclusion

Though there is a limitation with respect to the number of patients used in this study, this study made an attempt to specifically profile gene expression pattern of luminal B tumors of Indian women. We have used technical replicates and validated the expression profiles using TCGA data. This resulted in identification of 1692 genes commonly expressed between luminal B tumors of OU and TCGA cohorts. Further integrated analysis of OU and TCGA cohorts resulted in identification of 26 genes including 8 upregulated and 18 downregulated genes of potential prognostic value for luminal B tumors. Among the upregulated genes 7 genes are associated with DNA replication, repair and cell cycle-related pathways and among the downregulated genes 7 are associated with lipid metabolism-related pathways. We also identified differential splicing event in the prognostic gene ABCA10. Further in-depth analysis of these genes would result in identification of potential targets for luminal B breast cancer treatment.

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Author contributions KU was associated with planning, execution and manuscript preparation of the study. HSR was associated with planning the work and manuscript preparation and correction. KP and MS were associated with sample collection, processing and collection of clinical data. All authors read and approved the final manuscript.

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Declarations

Conflict of interest The authors declare that they have no financial or non-financial interests to declare.

Ethical approval This study was approved by Institutional Ethical Committee for Biomedical Research (NO.26/IEC/IOG/OU/18).

Consent to participate Written informed consent about the use of these samples was obtained from all patients included in this study.

Consent for publish Not applicable.

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